1. Part 1 Manual Calculations of K-Means Clustering

Note on notation:
A point is denoted as: P_i
A centroid is denoted as: C_i
The euclidean distance between two points: $euc(P_i, P_j)$

Formula for updating a Centroid, C_i
$C_i = \frac{P_1+P_2+...+P_k}{k}$

Formula for calculating euclidean distance between two points in 3D space
$euc(P_1, P_2) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$

Data:
Point 1 : $(1, 1, 1)$
Point 2 : $(1, 3, 1)$
Point 3 : $(2, 2, 2)$
Point 4 : $(4, 4, 4)$
Point 5 : $(4, 2, 2)$
Point 6 : $(3, 4, 2)$
Point 7 : $(3, 1, 2)$
Point 8 : $(2, 3, 1)$

Part 1: Assignments
Centroid 1: $C_1 = (1, 1, 1)$
Centroid 2: $C_2 = (4, 4, 4)$

Part 2: Calculations

Point $1 : (1, 1, 1)$
$euc(C_1, P_1) = \sqrt{(1-1)^2 + (1-1)^2 + (1-1)^2} = 0.00$
$euc(C_2, P_1) = \sqrt{(4-1)^2 + (4-1)^2 + (4-1)^2} = 5.20$
Point 1 is in C_1 as $0.00 < 5.20$

Point $2 : (1, 3, 1)$
$euc(C_1, P_2) = \sqrt{(1-1)^2 + (1-3)^2 + (1-3)^2} = 2.00$
$euc(C_2, P_2) = \sqrt{(4-1)^2 + (4-3)^2 + (4-1)^2} = 5.29$
Point 2 is in C_1 as $2.00 < 5.29$

Point $3 : (2, 2, 2)$
$euc(C_1, P_3) = \sqrt{(1-2)^2 + (1-2)^2 + (1-2)^2} = 1.73$
$euc(C_2, P_3) = \sqrt{(4-2)^2 + (4-2)^2 + (4-2)^2} = 2.45$
Point 3 is in C_1 as $1.73 < 2.45$

Point $4 : (4, 4, 4)$
$euc(C_1, P_4) = \sqrt{(1-4)^2 + (1-4)^2 + (1-4)^2} = 5.20$
$euc(C_2, P_4) = \sqrt{(4-4)^2 + (4-4)^2 + (4-4)^2} = 0.00$
Point 4 is in C_2 as $5.20 > 0.00$

Point $5 : (4, 2, 2)$
$euc(C_1, P_5) = \sqrt{(1-4)^2 + (1-2)^2 + (1-2)^2} = 3.32$
$euc(C_2, P_5) = \sqrt{(4-4)^2 + (4-2)^2 + (4-2)^2} = 2.82$
Point 5 is in C_2 as $3.32 > 2.00$

Point 6: (3, 4, 2)
$euc(C_1, P_6) = \sqrt{(1-3)^2 + (1-4)^2 + (1-2)^2} = 3.74$
$euc(C_2, P_6) = \sqrt{(4-3)^2 + (4-4)^2 + (4-2)^2} = 2.24$
Point 6 is in C_2 as $3.74 > 2.24$

Point 7: (3, 1, 2)
$euc(C_1, P_7) = \sqrt{(1-3)^2 + (1-1)^2 + (1-2)^2} = 2.24$
$euc(C_2, P_7) = \sqrt{(4-3)^2 + (4-1)^2 + (4-2)^2} = 3.74$
Point 7 is in C_1 as $2.24 < 3.74$

Point 8: (2, 3, 1)
$euc(C_1, P_8) = \sqrt{(1-2)^2 + (1-3)^2 + (1-1)^2} = 2.24$
$euc(C_2, P_8) = \sqrt{(4-2)^2 + (4-3)^2 + (4-1)^2} = 3.74$
Point 8 is in C_1 as $2.24 < 3.74$

Classifications before Updating Centroids:
Cluster 1 = $\{P_1, P_2, P_3, P_7, P_8\}$
Cluster 2 = $\{P_4, P_5, P_6\}$

Part 3: Updating Centroids and Reclassification with Mean Distances

Iteration 1
Step 1: Recalculate Centroids:
$C_1 = \frac{P_1+P_2+P_3+P_7+P_8}{4} = (\frac{1+1+2+3+2}{5}, \frac{1+3+2+1+3}{5}, \frac{1+1+2+2+1}{5}) = (1.8, 2.0, 1.4)$
$C_2 = \frac{P_4+P_5+P_6}{3} = (\frac{4+4+3}{3}, \frac{4+2+4}{3}, \frac{2+4+2}{3}) = (3.67, 3.33, 2.67)$

Reorganizing points:

Point 1: $(1, 1, 1)$
$euc(C_1, P_1) = \sqrt{(1.8-1)^2 + (2.0-1)^2 + (1.4-1)^2} = 1.34$
$euc(C_2, P_1) = \sqrt{(3.67-1)^2 + (3.33-1)^2 + (2.67-1)^2} = 3.92$
Point 1 does not change classifications as $1.34 < 3.92$

Point 2: $(1, 3, 1)$
$euc(C_1, P_2) = \sqrt{(1.8-1)^2 + (2.0-3)^2 + (1.4-3)^2} = 2.05$
$euc(C_2, P_2) = \sqrt{(3.67-1)^2 + (3.33-3)^2 + (2.67-1)^2} = 3.17$
Point 2 does not change classifications as $2.05 < 3.17$

Point 3: $(2, 2, 2)$
$euc(C_1, P_3) = \sqrt{(1.8-2)^2 + (2.0-2)^2 + (1.4-2)^2} = 0.63$
$euc(C_2, P_3) = \sqrt{(3.67-2)^2 + (3.33-2)^2 + (2.67-2)^2} = 2.71$
Point 3 does not change classifications as $0.63 < 2.71$

Point 4: $(4, 4, 4)$
$euc(C_1, P_4) = \sqrt{(1.8-4)^2 + (2.0-4)^2 + (1.4-4)^2} = 3.95$
$euc(C_2, P_4) = \sqrt{(3.67-4)^2 + (3.33-4)^2 + (2.67-4)^2} = 1.53$
Point 4 does not change classifications as $3.95 > 1.53$

Point 5: $(4, 2, 2)$
$euc(C_1, P_5) = \sqrt{(1.8-4)^2 + (2.0-2)^2 + (1.4-2)^2} = 2.28$
$euc(C_2, P_5) = \sqrt{(3.67-4)^2 + (3.33-2)^2 + (2.67-2)^2} = 1.52$
Point 5 does not change classifications as $2.28 > 1.52$

Point 6: $(3, 4, 2)$
$euc(C_1, P_6) = \sqrt{(1.8-3)^2 + (2.0-4)^2 + (1.4-2)^2} = 1.70$
$euc(C_2, P_6) = \sqrt{(3.67-3)^2 + (3.33-4)^2 + (2.67-2)^2} = 1.16$
Point 6 does not change classifications as $1.70 > 1.16$

Point 7:$(3, 1, 2)$
$euc(C_1, P_7) = \sqrt{(1.8-3)^2 + (2.0-1)^2 + (1.4-2)^2} = 2.41$
$euc(C_2, P_7) = \sqrt{(3.67-3)^2 + (3.33-1)^2 + (2.67-2)^2} = 2.52$
Point 7 does not change classifications as $2.41 < 2.52$

Point 8: $(2, 3, 1)$
$euc(C_1, P_8) = \sqrt{(1.8-2)^2 + (2.0-3)^2 + (1.4-1)^2} = 1.10$
$euc(C_2, P_8) = \sqrt{(3.67-2)^2 + (3.33-3)^2 + (2.67-1)^2} = 2.38$
Point 8 does not change classifications as $1.10 < 2.38$

Results: No points were reclassified, therefore no more iterations are to be made.

$C_1 = (1.8, 2.0, 1.4)$, Cluster 1 $= \{P_1, P_2, P_3, P_7, P_8\}$
$C_2 = (3.67, 3.33, 2.67)$, Cluster 2 $= \{P_4, P_5, P_6\}$

Adding new points to the model:
Point 9: $(2, 1, 2)$
Point 10: $(1, 1, 2)$
Point 11: $(3, 3, 3)$

Point 9: $(2, 1, 2)$
$euc(C_1, P_9) = \sqrt{(1.8-2)^2 + (2.0-1)^2 + (1.4-2)^2} = 1.18$
$euc(C_2, P_9) = \sqrt{(3.67-2)^2 + (3.33-1)^2 + (2.67-2)^2} = 2.94$
Point 9 is added to C_1

Point 10: $(1, 1, 2)$
$euc(C_1, P_10) = \sqrt{(1.8-1)^2 + (2.0-1)^2 + (1.4-2)^2} = 1.41$
$euc(C_2, P_10) = \sqrt{(3.67-1)^2 + (3.33-1)^2 + (2.67-2)^2} = 3.61$

Point 10 is added to C_1

Point 11: $(3, 3, 3)$
$euc(C_1, P_11) = \sqrt{(1.8 - 3)^2 + (2.0 - 3)^2 + (1.4 - 3)^2} = 2.24$
$euc(C_2, P_11) = \sqrt{(3.67 - 3)^2 + (3.33 - 3)^2 + (2.67 - 3)^2} = 0.82$
Point 11 is added to C_2